

CARDSTON, AB, Canada

WMO: 711530

Lat: 49.201N

Lon: 113.286W

Elev: 3727

StdP: 12.82

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-11.3	-23.6	1.7	-16.9	-17.0	2.5	-10.5	28.4	43.7	25.8	42.3	4.4	220	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	87.5	61.3	84.1	60.5	80.5	59.3	64.9	79.2	63.0	77.9	61.3	76.4	7.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.4	90.2	70.7	57.9	82.3	68.0	55.6	75.6	65.9	32.0	78.9	30.4	78.0	29.1	76.5	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.0	20.1	17.8	DB	-26.9	92.9	6.9	3.8	-31.8	95.7	-35.8	97.9	-39.7	100.0	-44.7	102.8	(4)
(5)			WB	-27.1	69.2	6.8	2.6	-32.0	71.1	-36.0	72.6	-39.8	74.1	-44.7	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	23.9	24.6	31.2	40.5	49.7	56.9	64.4	63.2	54.4	42.8	31.9	23.9		
	DBStd	18.34	16.15	14.59	13.55	9.03	8.19	6.24	5.91	6.27	8.30	9.50	13.46	14.69		
	HDD50	4189	810	711	585	301	106	12	1	2	49	255	549	808		
	HDD65	8389	1274	1130	1048	735	476	252	83	108	327	689	994	1273		
	CDD50	1410	1	0	2	16	98	219	447	411	180	31	5	0		
	CDD65	138	0	0	0	0	3	10	64	52	9	0	0	0		
	CDH74	2728	0	0	0	4	78	192	1103	1050	294	7	0	0		
	CDH80	842	0	0	0	0	12	36	376	351	67	0	0	0		
	WSAvg	7.7	8.5	7.9	8.2	8.4	8.1	7.5	6.5	6.1	6.6	7.9	8.3	8.3		
	PrecAvg	21.6	1.1	0.9	1.3	1.9	3.0	4.2	1.9	1.8	1.9	1.3	1.2	1.1		
(15) Precipitation	PrecMax	33.2	1.7	1.9	2.5	3.7	6.5	12.7	6.8	4.6	5.3	3.1	2.4	2.3		
	PrecMin	13.5	0.3	0.2	0.5	0.7	0.4	0.9	0.1	0.1	0.2	0.2	0.4	0.4		
	PrecStd	4.7	0.4	0.4	0.5	0.7	1.5	2.6	1.5	1.1	1.1	0.7	0.5	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.9	56.4	61.6	71.9	81.0	85.1	92.3	91.7	86.7	73.7	63.5	53.9		
		MCWB	42.5	41.4	44.9	51.7	58.0	63.8	63.3	61.2	58.5	53.1	47.3	40.5		
	2%	DB	50.1	50.9	56.4	65.8	75.5	79.3	88.1	87.4	81.3	68.0	57.4	49.4		
		MCWB	39.6	39.7	43.4	48.3	54.9	59.6	62.5	60.5	56.7	50.6	44.9	39.2		
	5%	DB	46.7	46.8	52.5	60.8	70.8	75.3	84.3	84.0	77.2	62.6	53.0	45.9		
		MCWB	38.0	37.7	41.6	46.2	53.0	58.0	61.8	59.5	55.9	48.7	42.3	37.5		
	10%	DB	43.7	43.3	48.6	55.8	66.0	71.2	80.3	80.3	71.9	57.8	49.0	42.5		
		MCWB	36.4	35.3	39.0	43.6	50.8	55.8	60.8	59.0	54.1	45.9	40.0	35.6		
	0.4%	WB	44.9	42.9	46.8	52.4	60.2	66.3	69.3	65.6	61.0	55.2	49.3	43.6		
		MCDB	53.3	53.4	58.6	69.1	77.8	78.2	84.2	79.8	79.2	70.5	60.4	51.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.0	40.6	44.4	49.4	56.3	62.1	66.3	63.5	58.9	52.0	45.7	40.3		
		MCDB	48.2	49.7	54.5	63.5	71.8	75.7	79.7	79.4	76.4	65.9	56.3	47.5		
	5%	WB	38.5	38.2	42.0	46.7	53.9	59.5	64.0	61.9	56.7	49.2	42.9	37.9		
		MCDB	45.9	46.6	51.6	59.2	68.4	71.7	78.4	78.1	73.4	61.2	52.2	44.8		
	10%	WB	36.5	35.5	39.5	44.2	51.8	57.2	62.2	60.3	54.7	46.6	40.3	35.5		
		MCDB	43.5	42.9	48.1	55.0	64.0	68.7	76.6	76.7	70.3	57.2	48.4	42.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.0	20.5	20.1	20.5	21.5	21.4	26.4	27.7	24.7	21.1	18.7	19.2		
		MCDBR	19.6	22.7	24.0	28.9	30.2	29.5	33.4	35.8	35.6	28.7	22.4	20.2		
		MCWBR	13.0	14.5	14.2	15.2	14.4	14.4	14.2	14.5	15.8	15.3	13.1	13.4		
		MCDBR	19.6	22.6	23.3	27.0	27.3	26.4	29.6	30.8	32.3	27.3	21.6	20.0		
	5% WB	MCWBR	13.8	14.9	14.4	15.1	13.5	13.9	14.6	13.7	15.1	15.1	13.2	14.2		
(40) Clear-Sky Solar Irradiance	taub		0.243	0.257	0.276	0.322	0.330	0.326	0.332	0.348	0.311	0.279	0.255	0.231		
	taud		2.408	2.407	2.453	2.361	2.413	2.470	2.458	2.398	2.475	2.544	2.498	2.419		
	Ebn at Noon		267	290	301	294	295	296	292	282	284	278	262	258		
	Edh at Noon		20	25	29	36	35	34	33	34	28	22	18	17		
(44) All-Sky Solar Radiation	RadAvg		393	711	1100	1519	1809	2023	2255	1803	1282	792	440	315		
	RadStd		32	53	97	99	159	157	126	115	105	77	33	29		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page